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US's orbital weapons: The limits of global pacts governing space militarisation

Amिताabh Sinha

New Delhi, September 15

THE US on Tuesday said it had active weapons deployed in space, the first such public admission by any country, sparking a fresh debate. The US did not reveal the nature of weapons deployed, or when they were deployed.

"Today we continue to ensure we remain ready to meet the challenges of evolving threats wherever they exist. This is why the US now has on-orbit space control weapons capable of defending the joint force against hostile adversary action," US Air Force Secretary Troy Meink said at a conference in Maryland, according to CNN.

There was immediate reaction from China and Russia, two countries that the US considers its space adversaries. "We urge the US side to stop expanding its military capabilities and preparing for war in outer space," Chinese foreign ministry spokesman Guo Jiakun told reporters, according to AFP.

That same report also quoted Kremlin spokesman Dmitry Peskov calling for keeping space free of any weapons. "We are counting on broad international consolidation to continue working toward the com-

plete demilitarisation of space," he said.

Interestingly, the US accuses China and Russia of improving their space capabilities with the objective of tracking and targeting US forces. "China and Russia are also testing and fielding sophisticated counter-space capabilities with the intent to disrupt and degrade US space-enabled capabilities," reads a factsheet on space threats, published on the website of the US Space Force, set up in 2019.

Weaponisation of space

Although this is the first time a country has publicly admitted to deploying weapons in space, the capabilities to do so are neither new nor a revelation. The military use of outer space covers a wide variety of capabilities: the ability to launch a missile from Earth to destroy enemy satellites or other space-based assets, the ability to send weapons from space to destroy ground-based targets, or the use of space-based weapons to target space-based assets. It is not clear which of these categories the US's newly-acknowledged weapons fall into.

But the weapons need not be "kinetic" — they need not necessarily be able to cause physical destruction of enemy targets. The ability to disrupt or jam the

★ STAR WARS

Use of weapons in space can take different forms. They can be used to either

● **destroy**, or to ♦ **jam**, equipment.

EARTH-TO-SPACE	SPACE-TO-SPACE	SPACE-TO-EARTH
Anti-satellite tests	Co-orbital satellites	Weapons deployed in space
● Involves launching missiles from earth to destroy space-based assets	One satellite can be programmed to crash into an enemy satellite	Satellite-based weapons can deorbit, enter atmosphere, and strike Earth-based targets
● Four countries demonstrated this	Soviet Union reported to have tested this during the Cold War	Not demonstrated
♦ Earth-based systems can jam or 'blind' signals of enemy satellites	Using space-based assets to jam or block the signals or comms of enemy satellites	Use of space-based systems to interfere or disrupt functioning of radars, missiles or aircraft
♦ Many have this capability	No known capability	Not known

enemy's communications between its space and Earth-based systems, or attack their cyber-networks, can also be extremely effective in a conflict situation.

Several countries have developed some, or all, of these capabilities, though not every such capability has been tested and

demonstrated.

The most open demonstrations of such capabilities have been the anti-satellite tests, in which countries send a missile to destroy a satellite in space. Four countries — the US, Russia, China and India — have demonstrated this. Though the countries

targeted their own non-functional satellites during the tests, this ability to completely destroy any space-based asset happens to be the most dramatic use of weapons in space to date.

There have been examples of non-kinetic methods as well. Just before the Russian attack on Ukraine in February 2022, Russian hackers gained control over ground stations of the Viasat satellite that was being used to provide internet services to Ukrainian subscribers, including their military agencies. There are also reports of attempts by Russia to block GPS signals in Ukraine.

Legal ambiguity

All countries emphasise on peaceful uses of space and call upon each other not to militarise space. But there is no international law that explicitly prohibits the deployment or use of such weapons or tactics in space. The Outer Space Treaty of 1967, an old but still the most relevant international law on this subject, bars countries from carrying or placing nuclear weapons or "other kinds of weapons of mass destruction" in space, but it is silent on conventional weapons, or those that target physical infrastructure in space. It does not say

anything about earth-to-space weapons, the development of which was still some distance away in the 1960s.

Attempts to create an international legal framework to regulate the deployment and use of weapons in space were made, but support from all the major space powers was lacking. Around 2008, China and Russia had together proposed the idea of a "Prevention of the Placement of Weapons in Outer Space" treaty banning the deployment of all weapons — not just nuclear and other weapons of mass destruction — in space, but this too left out earth-to-space weapons. This idea never came to fruition.

An ongoing discussion on "Prevention of an Arms Race in Outer Space" at the UN Conference on Disarmament has also not led to any international law or treaty.

Meanwhile, the Artemis Accords, an initiative of NASA and the US State Department, is a voluntary code of conduct on exploration of space. It has over 70 signatory countries, including India. All cooperative activities are meant to be for peaceful purposes, but there is nothing that prevents signatories from deploying or using weapons in space.

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Outer space Militarisation- "on-orbit space control weapons," USA

Earth to space , space to space, space to earth

Kinetic and non-kinetic (GPS/SATCOM jamming)

Outer space treaty of 1967- bars countries from

carrying or placing nuclear weapons or other kind

of weapons of mass destruction in space

Discussions on Prevention of an arms race in

outer space- UN Conference on Disarmament

Peaceful uses of space

Artemis Accords- NASA & US State department

Voluntary code of conduct on space exploration

70 signatory countries including India

As AI takes over more of our thinking, an MIT report warns of ‘cognitive surrender’

Mark Arsenault, Dana Goldstein, Alan Blinder & Sarah Mervosh
September 15

THE WORRIES that artificial intelligence is degrading education have been building for months. Students are using it for everything, including cheating, and a recent report out of MIT said it is triggering “cognitive surrender.” Many university leaders are embracing AI with enthusiasm anyway.

As AI leaders warn about the dangers of their own technology, the dissonance between the optimism of university leaders and the fears of many students and professors may increase. Profes-

sors and students are often in a bind about what they should do: Use more AI to stay relevant, or dial back to preserve learning and thinking?

While some politicians and AI leaders are warning about possible doomsday scenarios for humanity as AI models advance, many campus leaders are focused on existential threat the technology poses to universities.

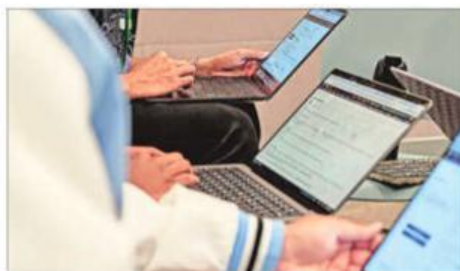
The committee at the Massachusetts Institute of Technology studying AI use released a number of alarming findings, including that the technology has caused “major shifts in campus culture,” isolating students who are skipping office hours and ditching communal studying.

Getting answers from a bot can create “the illusion of learning” for students who reach for AI “at the first hint of struggle,” the MIT committee wrote in its August report.

At the same time, “the potential of these technologies to augment work across campus is immense,” the committee wrote.

The MIT report crystallized some of the thinking in higher education about AI, but top administrators have been embroiled in debates for several years. Administrators have often been the most eager to champion AI efforts, while faculty and students remain skeptical.

“I think you have to lean in, but you have to do so in a way



Some varsity leaders say schools must embrace AI to stay relevant. NYT

that’s thoughtful and ethical and asks the right questions,” said Wendy Hensel, the president of the University of Hawaii, where

she presented an AI strategy to regents in July. “The right answer to AI is not just ‘no.’ It can’t be. It has to be approached

with curiosity and understanding of the implications.”

The hodgepodge of policies universities have created to respond to AI often reflect the confusion about how to proceed — with some clear dos and don’ts, but often a lot of wiggle room for faculty and students to figure it out on their own.

For a motivated student, an AI chatbot can function as a sort of tutor. Chatbots can break down the steps in a math problem, surface unexpected sources for research papers and offer feedback on writing.

Still, a growing body of research suggests that AI use can also harm learning.

A recent study of 26,000 Chi-

nese secondary-school students found that AI adoption led to an 18 percent increase in homework scores, but a 20 percent decline in scores on secure exams. The implication was that rushing through homework with help from chatbots degraded the deeper learning needed to perform well when the technology was not available.

One 2025 study out of MIT found that people who used chatbots as writing assistants struggled to remember what they had written only minutes after finishing their essay. “You basically didn’t engage any of the memory networks that would actually help you remember,” said Nataliya Kos-

myrna, a MIT researcher who worked on the study.

Other research has shown that people who have greater trust in AI and rely on it more are more likely to have lower cognitive engagement. But AI can be hard to avoid. Chatbots are embedded within the Google Classroom suite, which is nearly ubiquitous in schools, and within many online learning systems used by universities.

Some tools block outright cheating or plagiarism and allow professors to toggle specific AI capabilities on and off. But the most popular chat bots among students will complete entire assignments, including research papers, on student’s behalf. NYT

Cognitive surrender- Illusion of learning

AI improves the product, not necessarily the learner

Major shifts in campus culture- isolating students, skipping office hours and ditching communal studying

MIT recommends **rethinking assessment, strengthening hands-on learning, and establishing explicit AI-use rules for individual courses**, rather than imposing one universal policy.

“**answer provider**” to “**cognitive scaffold.**” The objective should be to make the student think *better*, not think *less*.

Peak water: a flow U-turn

Reuters

The Himalaya are approaching a tipping point as glaciers melt faster than a decade ago, threatening water security as the region approaches “peak water” by mid-century, according to a study released this month.

The findings come after the collapse of a Himalayan glacier in late August along the Nepal-Tibet border, which caused cascading landslides and flash floods in the valleys below. At least 1,300 people have been confirmed dead and more than 5,300 are missing across Nepal and China's Tibet region.

As glaciers retreat, they are leaving behind unstable glacial lakes held back by little more than loose rock and ice, above valleys where millions of people live, the study found.

Peak water is the point in time when river flows stop rising and start declining. It is expected to have profound implications for water security. As the Himalaya underpin more than 20% of India's GDP, the region makes up a key piece of the country's national economic infrastructure, upon which hundreds of millions of people depend.

The study was produced by Systemiq, a sustainability advisory firm, together with the Integrated Mountain Initiative, the International Centre for Integrated Mountain Development, and India's G.B. Pant National Institute of Himalayan Environment.



Tibetan peak: Snow-covered peaks of the Himalaya are visible in Shigatse, Tibet Autonomous Region, China. REUTERS

Himalayan glacier mass loss had accelerated in recent decades, while only 21 glaciers were currently monitored on the ground out of an estimated 40,000 glaciers across the Hindu Kush-Himalaya region, a vast mountain system spanning eight countries from Afghanistan to Myanmar, the study found.

Though making up around 18% of India's land, the Himalayan region accounted for roughly 35% of the country's natural disasters.

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Himalayan glaciers approaching 'peak water' by mid-century threaten long-term water security despite short-term flow increases from accelerated melting

What is “peak water”?

Peak water is the point at which a glacier's meltwater contribution to a river reaches its **maximum**. After that, continued warming and glacier shrinkage eventually cause the glacier to lose its capacity to sustain the same level of meltwater contribution

Wider implications for India

Agriculture: Irrigation systems

Hydropower: Increased runoff can temporarily raise generation potential, while later declining flows and greater hydrological variability

Disaster management: Rapid melting can contribute to **glacial lake formation and GLOF risks**, while extreme precipitation can compound hazards.

Groundwater: Reduced reliability of surface-water flows could increase dependence on groundwater, particularly in downstream agricultural regions.

Inter-state water relations:

What about the father's leave? It is parenting, not just mothering



THIAMIZHACHI THIAGARANDIAN

ON AUGUST 18, Tamil Nadu's Human Resources Management Minister D Sarathkumar announced that women government employees would receive over 365 days of maternity leave for the birth of a third child, up from the earlier 12 weeks. The state government takes it a step towards women's empowerment and social welfare, and framed it as a response to the state's declining fertility rate. On the surface, this is a welcome correction. The Madras High Court had also observed that the physical and caregiving demands of children should not diminish because a woman is having her third child. Twelve weeks was over enough, and removing that arbitrary cap was overdue.

Yet, I find myself returning to a question this scheme cannot answer, one I suspect it was never meant to ask. Why does the father's leave never grow larger?

Tamil Nadu did not arrive at this question recently. It was Periyar's Self-Respect Movement that insisted that marriage and motherhood were matters of women's dignity and consent, not obligations owed to custom, or family. It shaped how Dravidar Munnetra Kazhagam (DMK) governments approached women's welfare—not as a single generous gesture but as sustained, structural change. The Kalaignar government gave daughters equal rights to ancestral property. It was before the Union government followed in 2006. Free bus travel widened where a woman could go without needing a man's permission or her family's name change. The Kalaignar Magalir Ormai Thoppai placed a monthly aim directly into women's hands, recognising that independence requires economic control.

Historian Linda Gordon has argued that we cannot explain, or correct, women's oppression by conjoining of reproduction from how work, wages and time are organised for everyone else. Tamil Nadu's maternity leave scheme does precisely this. By speaking only in the language of welfare and care, it prevents the laid-back division of parental responsibility as a natural difference between mothers and fathers.

Sweden offers a useful lesson here. In 1974, it became the first country to replace maternity leave with a gender-neutral parental leave benefit, allowing parents to decide between them. However, fathers' participation remained low for decades because leave could simply be transferred to mothers. Sweden eventually introduced non-transferable leave reserved for each parent. The lesson is not that fathers should be given leave, but that policy has to be designed in a way that makes caregiving a shared responsibility.



We have not even begun this conversation. We do not have a national paternity leave policy; the name, let alone one built on the understanding that raising a child is not a woman's assignment with occasional male assistance.

This brings me to the second, more uncomfortable part of the government's own justification: That this measure responds to a declining fertility rate. But extending maternity leave does not address why fertility is declining in the first place. Across the world, women are choosing to have fewer children because the economic and social cost of raising a child has become harder to justify. If we are serious about reversing declining birth rates, the answer is paid, mandatory, non-transferable leave for fathers.

I am reminded of Betty Friedan's *The Feminine Mystique* (1963) and how carefully, in the postwar years, domesticity was repeatedly sold to women as fulfillment rather than constraint. There is a version of that same story being retold today, dressed in the language of welfare and women's empowerment.

There is a question this scheme does not ask. A year of paid leave tied to a third pregnancy does not exist in a vacuum. A scheme that pushes women towards childbearing risks placing marginalised women in vulnerable positions while failing to address the larger question of care. A woman's body, and what she chooses to do with it, is not a lever for the state to pull in service of a fertility target.

This scheme, then, is not the full response to declining birth rates and the unequal burden of care it presents itself as. Every woman who benefits from this scheme deserves that time, and more. But this is the full extent of the state's response. It tells us something about where Indian society still places responsibility for the next generation, regardless of who is in government. Until that changes, adding more days to maternity leave will not solve the problem it claims to address.

365 days of maternity leave for 3rd child (TN) from earlier 12 weeks.

Response to declining fertility rate

Care giving a shared responsibility

Periyar's self respect movement- " marriage and motherhood were matters of a woman's dignity and consent, not obligations owed to custom, or family."

Raising a child is not a woman's assignment with occasional male assistant

Need for a National Paternity leave policy (Now 15 days for central government employee)

Sweden model

non-transferable, 'use-it-or-lose-it' parental leave for fathers in addressing

declining fertility rates and long-term gender equity in caregiving

Moves from "motherhood" to "parenthood"

Decoding India's GDP base revision

Changing the GDP base year is not just about changing how we calculate economic growth. It can also change our estimate of how large the economy was in rupee terms – what is known as nominal GDP. When one gets a better picture of the economy through new data, improved methods and wider coverage, our understanding of the economy has to be reviewed. Importantly, as recognised in international statistical practice, such revisions do not necessarily move in one direction: the estimated size of the economy can go up or down, depending on what new data, methods reveal.

With the recent base revision of GDP, across the three years for which both series overlap, India's nominal GDP was revised down by about 2.7% in 2022-23, 3.5% in 2023-24 and 3.8% during 2024-25. However, India is not alone in seeing nominal GDP estimates change when the GDP base is revised. Rebasings exercises in countries such as Nigeria (2014), Indonesia (2014), Brazil (2015), South Africa (2018), Mexico (2019), China (2021) and Spain (2024) all resulted in revisions to previously estimated nominal GDP levels. India's earlier shift from base year 2004-05 to the 2011-12 also changed the estimated size of the economy. The important question, therefore, is not simply why the number changed, but what changed in our data or methods which led to the new number.

It is worthwhile to mention that the Press Release dated February 27, 2026 on the New GDP Series



Saibal Chattopadhyay

Chairman, National Statistical Commission



Saurabh Garg

Secretary, Ministry of Statistics and Programme Implementation

with base year 2022-23 had already indicated the various methodological improvements and updated data sources which contributed to revisions in the nominal GDP, along with a comparative table indicating economic activity wise revisions and the reasons for such revisions.

Better evidence

Much of the media attention has focused on the headline downward revision in GDP, while overlooking the substantial upward revisions in several sectors. A proper assessment requires looking at the sector-wise picture, where the revisions are far from uniform. Agriculture and allied activities were revised up by about 3.8%-5.9%, while financial services and real estate, professional services and ownership of dwellings were revised up by roughly 7.8%-9.0% over comparable years. In contrast, trade and transport and storage saw sharp downward revisions of around 23%-26%. Trade GVA (Gross Value Added) was revised downwards by 36%, road transport by 16.9%, whereas hotels and restaurants' GVA was revised upwards by 5.7%, mainly on account of the revised estimates of the unincorporated sector. In the 2011-12 series, this sector was estimated by moving forward benchmark estimates with proxy indicators whereas in the new series, ASUSE (Annual Survey of Unincorporated Sector Enterprises) and PLFS (Periodic Labour Force Survey) data provided a more direct basis for measurement of the unincorporated sector.

Therefore, a major driver of the revision in nominal GDP in the case of India is

the improved measurement of India's unincorporated services sector. The World Bank's April 2026 India Development Update similarly notes that India's nominal GDP was revised down by 3-4% in each of the four years from FY23, mainly due to a reassessment of the informal economy. It also notes that the revised series shows quarterly growth between FY 2023-24 to FY 2025-26 to be less volatile and more broad-based than previously estimated. However, it has to be noted that revisions in the unincorporated sector have not been uniformly revised but vary sector to sector. Moreover, India's quarterly and provisional GDP estimates are built from the previous year's quarterly figures and then updated using information such as GST collections and industrial production. So, when the FY 2022-23 GDP estimate was revised under the new methodology, the change naturally carried forward into the years that followed and all subsequent annual and quarterly estimates moved down with it. This does not mean the economy suddenly became smaller or slowed down in those years; part of the change reflects that we are now starting the calculation from a different, better-measured number.

Ultimately, the new GDP series should be judged not by a single headline revision, but by the effort to make the measurement of the Indian economy more current. The real test of a statistical system is not whether its numbers remain unchanged, but whether it is willing to change them when better evidence becomes available.

GS-3 Economy - National Income Accounting

27 Feb 2026, new GDP Series base year revision , 2022-23
Sector-wise picture

Agriculture and Allied activities up by 3.8%-5.9%

Financial Services, Real estate professional services and ownership of dwellings up by 7.8%-9%

Trade, transport and storage down by 23%-26%
(Trade GVA -36%, Road transport -16.9%)

Hotels and restaurants GVA up 5.7%

India's Nominal GDP down by 3-4% from FY23

Improved measurement of India's unincorporated service sector
Reassessment of the informal economy

The new 2022–23 series uses:

- **ASUSE** provides information on output, value added, employment and other characteristics of **unincorporated** enterprises.

- **PLFS** provides detailed information on the workforce and employment structure.

India's economy has changed considerably since 2011–12, particularly through **formalisation, digitalisation, new service activities and changing labour-market structures.**

Kishau dam project inked: The long delay in project, stakes for 6 states

Saurabh Parashar & Harikishan Sharma
Shimla, New Delhi, September 15

THE CHIEF ministers of six states on Tuesday signed a memorandum of understanding (MoU) to pave the way for the implementation of the long-pending 422 MW Kishau Multipurpose Project.

Himachal Pradesh Chief Minister Sukhvinder Singh Sukhu signed the MoU for the Rs 15,000-crore project with Uttarakhand CM Pushkar Singh Dhami, Uttar Pradesh CM Yogi Adityanath, Haryana CM Nayab Singh Saini, Rajasthan CM Bhajan Lal Sharma, and Delhi CM Rekha Gupta. Union Home Minister Amit Shah and Union Jal Shakti Minister CR Patil were present. The Union Cabinet will now have to approve the project.

What is the Kishau Multipurpose Project?

The Kishau Multipurpose Project is a proposed dam and hydropower project on the Tons River, a major tributary of the Yamuna, along the border shared by Himachal Pradesh and Uttarakhand. The project envisages a 232.6-metre-high concrete gravity dam that will generate 1,476 million units of clean hydropower.

The project site straddles Himachal Pradesh's Shimla district and Uttarakhand's Dehradun district. The two states signed an MoU to form a joint venture in June 2015, and Kishau Corporation Ltd was incorporated in January 2017. The project is intended to generate electricity, create irrigation potential of about 97,076 hectares, and provide drinking and industrial water to the beneficiary states, Haryana, Delhi, Rajasthan and Uttar Pradesh.

Why was the Kishau Dam required?

Kishau Dam was one of three storage projects planned in the upper reaches of the Yamuna and its tributaries following a MoU signed on May 12, 1994 by the Yamuna basin States — undivided Uttar Pradesh, Haryana, Rajasthan, Himachal Pradesh, and the National Capital Territory of Delhi — regarding the allocation of Yamuna water.

The other two projects were Lakhwar and Renukaji, for which the MoUs have already been signed and are under implementation. These projects were required because there are no storage facilities constructed on the Upper Yamuna — from its source, Yamunotri glacier, in the lower Himalayas to the Okhla barrage in Delhi — a catchment area of about 30,000 square kilometres. As per the 1994 MoU, an interim seasonal allocation of the annual utilisable flow of river Yamuna was made among these five states.

Why was the signing of the MoU delayed?

For eight years, Himachal Pradesh held out on signing the MoU for two key reasons — the project cost burden, particularly the power component, and concerns that the state would not enjoy proportionate benefits.

Under an earlier arrangement, the previous BJP-led Himachal government agreed to contribute about Rs 800 crore towards the project cost. The current Congress-led government rejected that arrangement, saying it was not in Himachal's financial interest, and that it would suffer substantial land submergence and displacement.

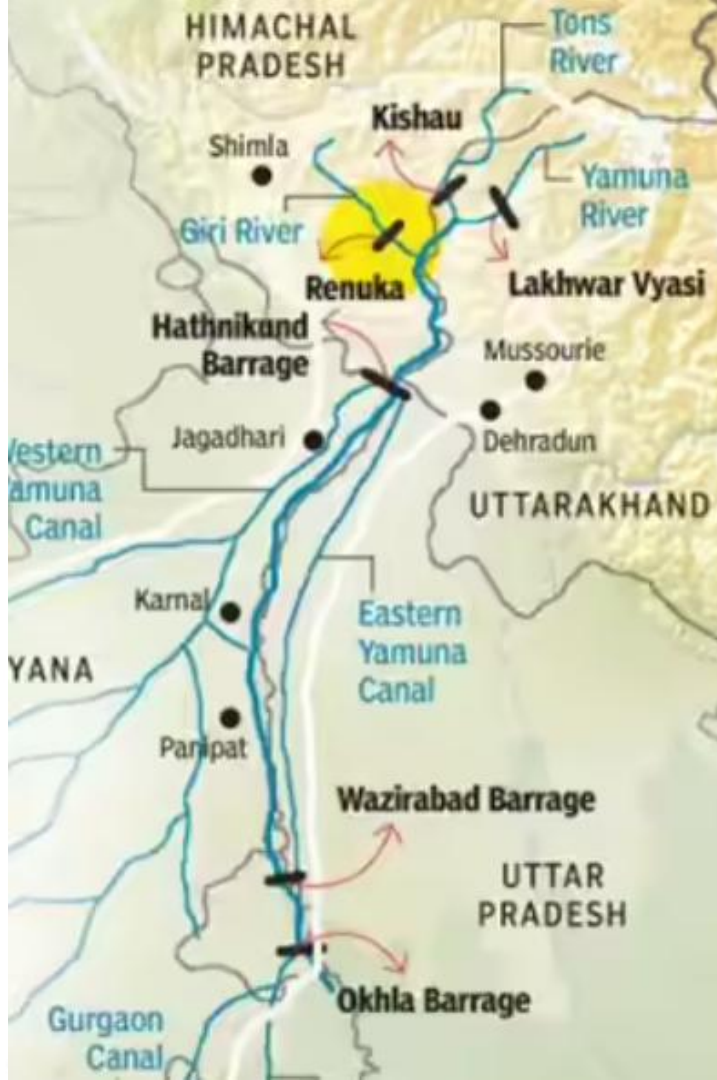
Indeed, the project will result in substantial submergence in both Himachal Pradesh and Uttarakhand.

According to a 2020 feasibility study, around 2,960 hectares of land would be submerged, affecting 17 villages and about 5,500 people, across the two states.

How was a solution reached?

A breakthrough was achieved in June 2026 when the Centre announced that it would bear 90% of the project cost, with the six states sharing the remaining 10%.

A senior Himachal Pradesh officer told *The Indian Express*: "This project will have a storage capacity of 1,786 million cubic metres (MCM) of water but will actually store 1,324 MCM. As per the joint agreements... Haryana will get 633 MCM; Uttar Pradesh will get 411 MCM; Rajasthan will get 124 MCM; Delhi will get 80 MCM; Himachal will get 42 MCM and Uttarakhand will get 34 MCM for drinking and industrial purposes."



What is the Kishau Project?

Kishau is a proposed **multipurpose storage dam on the Tons River**, on the Uttarakhand–Himachal Pradesh border. The project is intended to provide **water storage/irrigation and drinking-industrial water**, besides hydropower.

The six participating states are:

- Himachal Pradesh
- Uttarakhand
- Haryana
- Uttar Pradesh
- Rajasthan
- Delhi

Center 90% and States 10% cost sharing

- water benefits;
- hydropower benefits;
- irrigation;
- drinking water;
- project costs;
- environmental/social costs.

Thank you

“The price of success is daily effort”



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